

Subject-Normalized Stress-State Discovery from Wearable Biosignals

A baseline-aware representation study using HR, TEMP, and EDA features

KEY TAKEAWAY

Weak but nonzero stress signal: individual baselines dominate raw PCA, so preprocessing and LOSO validation are essential.

Problem & Data

WHAT IS BEING TESTED

312

OBSERVATIONS

26

INDIVIDUALS

51

FEATURES

HR

TEMP

EDA

wearable biosignals

- EmoPairCompete dataset with repeated rest, competition, and recovery phases.
- Question: do low-dimensional biosignal features capture phase, or mostly who was measured?
- Balanced phase design: 104 observations per phase.

Phase 1

pre-rest

Phase 2

puzzle competition

Phase 3

post-rest

Pipeline

PREPROCESS, REPRESENT, VALIDATE

01

Median imputation

Resolve sparse missing EDA features

02

Raw scaling

Standardize feature scale

03

Subject-wise z-score

Remove personal baseline offsets

04

Phase1 baseline subtraction

Model phase2/phase3 as deviations

05

PCA representation

Use PC1-PC5 for structure checks

06

KMeans clustering

k=3 compared against phase labels

07

LOSO sanity check

Supervised subject-level validation

Key Results

SIGNAL IS WEAK AND BASELINE-DEPENDENT

RAW PHASE R₂

3.0%

PC1-PC5

SUBJECT Z PHASE R₂

9.9%

improved

KMEANS ARI

0.075

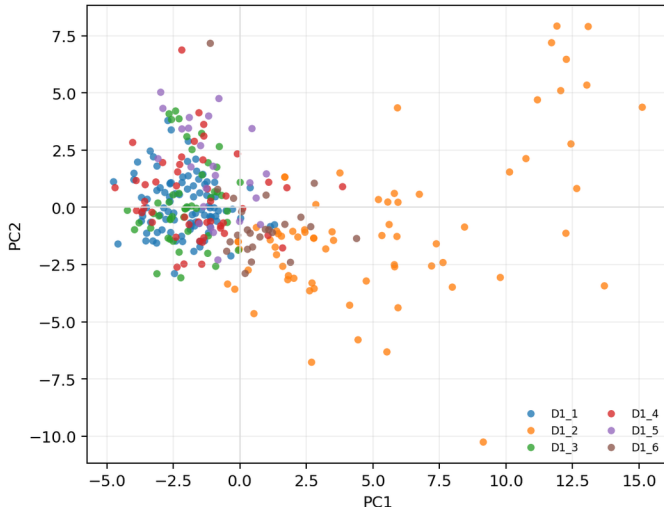
subject z

LOSO AUC

0.779

phase2 vs phase1

Raw standardized features



Subject-normalized features

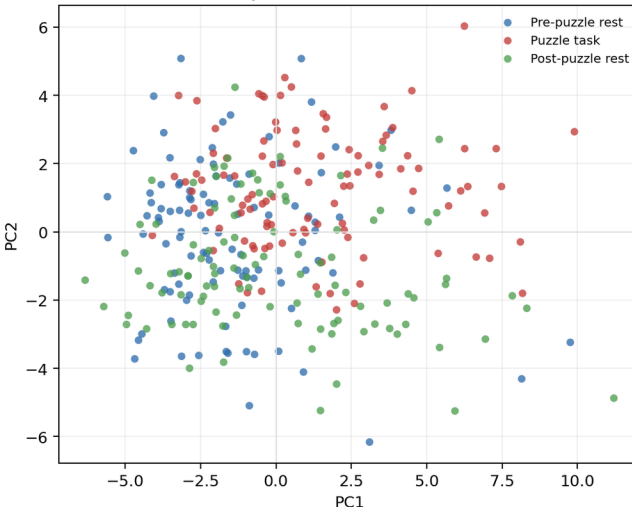


Figure 1. Raw PCA is visibly cohort/individual dominated; subject z-score reduces baseline shifts but phases still overlap.

Which metadata explain the low-dimensional representation?

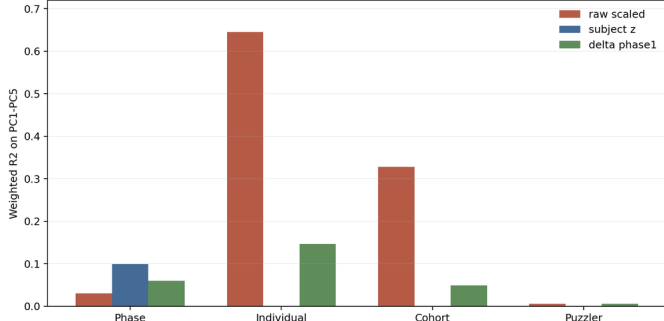


Figure 2. Metadata R₂ on PC1-PC5; raw individual 64.5%, cohort 32.8%, phase 3.0%.

KMeans clusters vs experimental phases

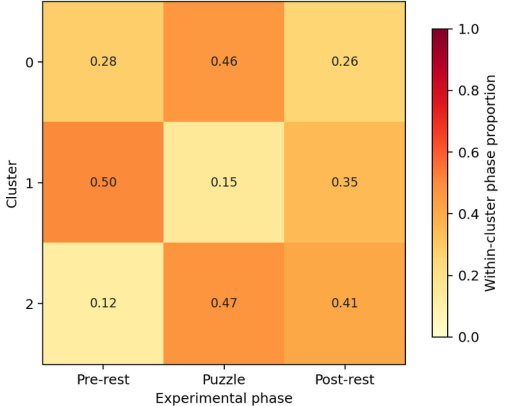


Figure 3. KMeans clusters after subject normalization are phase mixtures, not clean phase-pure states.

Result summary: PCA + KMeans does not recover clean phase clusters; supervision confirms only weak nonzero signal.

Interpretation

HOW TO READ THE NEGATIVE RESULT

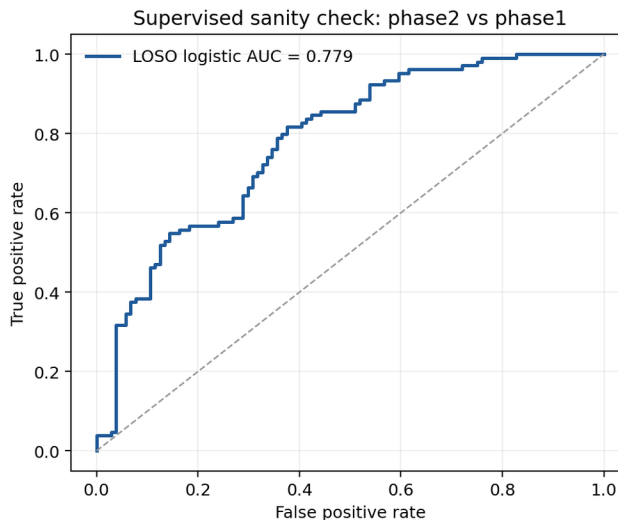


Figure 4. LOSO logistic regression reaches AUC about 0.779, so the data are not uninformative.

- Wearable biosignals contain weak but nonzero stress-related signal.
- Individual baselines dominate raw representations.
- Subject-wise normalization is necessary before interpretation.
- PCA + KMeans should not be overclaimed as discovering clean emotional states.
- Subject-level validation is essential for wearable biosignal claims.

Main conclusion

Baseline-aware preprocessing and subject-level validation are essential before interpreting wearable biosignals as invariant stress states.